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Stronger memory representation after memory reinstatement during retrieval in the human hippocampus

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Memory retrieval allows us to reinstate previously encoded information but is also considered to contribute to memory enhancement. Retrieval-induced enhancement may involve processing to strengthen memory traces, but neural processing beyond reinstatement during retrieval remains elusive. Here, we show that hippocampal processing, different from memory reinstatement, exists during retrieval in the human brain. By tracking changes in the response patterns in the selected hippocampal and cortical regions over time during retrieval based on functional MRI, we found that the representation of associative memory in CA3/DG became stronger even after cortical memory reinstatement, while CA1 showed significant memory representation at retrieval onset with the cortical reinstatement, but not afterwards. This tendency was not observed in the condition without active retrieval. Moreover, subsequent long-term memory performance depended on the delayed CA3/DG representation during retrieval. These findings suggest that CA3/DG contributes to neural processing beyond memory reinstatement during retrieval, which may lead to memory enhancement.

1. Introduction

Memory retrieval enables us to reinstate and utilize previously stored information that is relevant to the current situation. Traditionally, memory models have suggested that the hippocampus plays a critical role in the reinstatement of memory representations during retrieval (McClelland et al., 1995; Norman and O'Reilly, 2003; Teyler and DiScenna, 1986; Teyler and Rudy, 2007). These studies suggest that memory retrieval is initiated when a memory cue activates the distilled traces in the hippocampus, and is completed when the sensory details associated with the distilled traces are reinstated in the cortical areas through the interaction between the hippocampus and the cortical regions (Teyler and DiScenna, 1986; Teyler and Rudy, 2007). Neuroimaging studies support this idea. The reinstatement of information in the visual cortex (Bosch et al., 2014; Tanaka et al., 2014) or in the parahippocampal cortex (Staresina et al., 2012a; Tompary et al., 2016) depends on the level of hippocampal activity during retrieval. Moreover, based on the hippocampal responses during retrieval, the decoding of different episodic or associative memory contents was possible (Backus et al., 2016; Chadwick et al., 2010). However, beyond the memory reinstatement process, retrieval itself can act as a signal to change original memory traces. Memory reconsolidation studies propose that consolidated memory becomes labile again for a while after retrieval onset, allowing strengthening or modification of the original memory traces (Alberini and Ledoux, 2013; Besnard et al., 2012; Nader and Einarsson, 2010). Studies of a testing effect suggest that the retrieval signal itself can strengthen the retrieved memory traces even in the absence of any feedback or additional learning (Guran et al., 2020; Karpicke and Roediger, 2008; Roediger and Butler, 2011; Roediger and Challis, 1992). The hippocampus is also considered to be crucially involved in the strengthening or modification of memory induced by retrieval signals (Debiec et al., 2002; Forcato et al., 2016; Rossato et al., 2007; Schwabe et al., 2012). In particular, fMRI studies have shown that hippocampal activity levels increase during retrieval practice, with the hippocampal activity during retrieval also correlated with subsequent memory performance, supporting the involvement of the hippocampus in retrieval-induced memory enhancement (Jonker et al., 2018; Sekeres et al., 2021; Wiklund-Hörnqvist et al., 2021; Wing et al., 2013).

How does the human hippocampus handle the two processes of reinstatement and strengthening during retrieval? Memory reinstatement models suggest that memory cue information is conveyed to CA3, which is critical for hippocampal pattern completion through its recurrent connections, followed by the generation of neural representations in CA1 and full memory reinstatement in the cortical regions (Duncan et al., 2012; Granger et al., 1996; Lisman, 1999; McClelland and Goddard, 1996; Rolls, 1996; Tompary et al., 2016; Treves and

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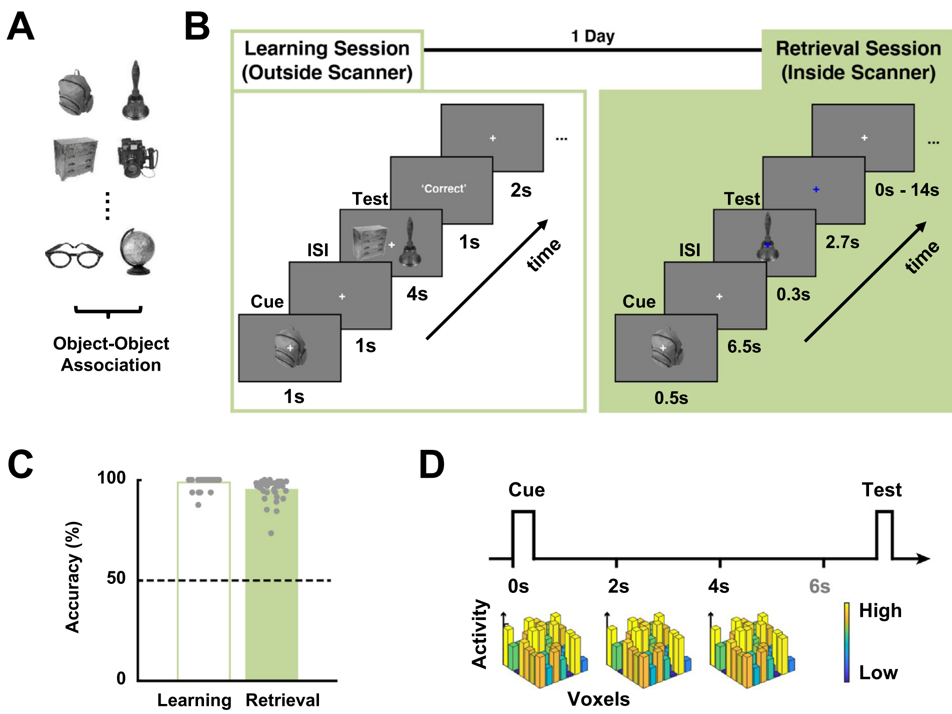

Fig. 1. Experimental design and memory performance. (A) Examples of object stimuli used in the main task. The task is based on object-object associations. **(B)** Participants performed an associative memory task, which consisted of Learning and Retrieval sessions. During the Learning session, the participants were trained to associate the eight object-object pairs. In each trial of this session, a cue object and two test objects were presented sequentially. They were instructed to choose one of the two test objects associated with the cue object, followed by feedback for the choice. On the following day, the participants completed the Retrieval session inside the MRI scanner. In each trial of this session, they were instructed to retrieve the associated object for a presented cue object and indicate whether the test object was associated with the cue object. **(C)** Retrieval accuracy on the last block of the Learning session (left bar) and retrieval accuracy during the Retrieval session (right bar). Each dot indicates accuracy for each participant. **(D)** The response patterns evoked at 0 s (retrieval onset), 2 s, and 4 s following retrieval onset were used to examine the change of associative memory representation during retrieval.

Rolls, 1994). Therefore, memory reinstatement is thought to occur in the direction from DG to CA3, CA1, and the cortex, in a very short range of time (500~1500ms; Staresina and Wimber, 2019). If memory retrieval signal itself can trigger the memory strengthening process in the hippocampal regions, this processing is likely to proceed in parallel with the reinstatement processing. Moreover, even if the reinstatement and the strengthening are triggered simultaneously, the time course of the strengthening process may be different from the rapid process of reinstatement. However, while detailed mechanisms have been proposed for memory reinstatement processing in the hippocampus, the hippocampal processing related to memory strengthening during retrieval remains unclear.

To investigate the changes in memory representations during retrieval, we tracked changes in brain activity with fMRI while participants retrieved associative object memories. To do this, participants were asked to conduct an associative memory task consisting of "Learning" and "Retrieval" sessions (Figs. 1A and 1B). In the Learning session, the participants were trained to learn object-object associations (Figs. 1A, 1B and S1A). On the following day, in the Retrieval session, the participants were asked to retrieve the associated object when one of the learned objects (cue object) was presented inside the scanner (Fig. 1B). We tracked how the associative memory representations changed in the hippocampal regions from the moment of retrieval onset (cue object onset) for the initial seconds of retrieval, using the estimated peak amplitudes of HRFs for the events at 0 s (retrieval onset), 2 s, and 4 s after retrieval onset in the representational similarity analysis (RSA) (Haxby et al., 2001; Kriegeskorte et al., 2008; Park et al., 2021) (Fig. 1D). By comparing these hippocampal representational changes over time during retrieval with memory reinstatement in the visual cortex, we examined the possibility of the existence of information processing beyond memory reinstatement in the hippocampus. Moreover, to confirm whether the altered memory representations in the hippocampal regions depend on the retrieval process, we also examined the hippocampal responses when the participants viewed the learned object without active retrieval. Furthermore, we examined whether memory performance long after (six months or more) retrieval depends on the change of memory representations during the retrieval in the hippocampus.

2. Material and methods

2.1. Participants

Thirty-eight neurologically healthy right-handed Korean participants (13 female and 25 male; mean age = 22.71 ± 1.87 years) took part in the fMRI experiment. All participants had normal or corrected to normal vision and reported no color blindness. Four additional participants were excluded because they showed low accuracy level in the Retrieval session (lower than 87.5%, implying potential loss of a pair out of a total of eight pairs). All participants provided a written informed consent for the procedure in accordance with protocols approved by the KAIST Institutional Review Board.

2.2. Stimuli and task design

We used 16 grayscale object (bag, bell, car, chair, glasses, globe, hat, medal, drawers, camera, clock, flag, glove, guitar, lantern, mask) images (Fig. S1A). Before the task, a pre-learning assessment was conducted to check if the participants had any prior knowledge about the relationships between the stimuli. In the pre-learning assessment, all possible pairs (120 pairs in total) between the 16 object images were presented sequentially and the participants were asked to grade the degree of relationship between two simultaneously presented objects on a five-point scale (1: not related at all, 2: does not seem to be related, 3: do not know, 4: slightly related, 5: highly related). The mean score was 1.93 ± 0.09 and this value was significantly lower than 3 ('do not know'), indicating no significant relationship between the object pairs. In the main task, we used eight pairs of unrelated objects.

The main task consisted of Learning and Retrieval sessions (Fig. 1B). The Learning session was conducted in a testing room outside the scanner, while the Retrieval session was conducted inside the MRI scanner one day after the Learning session. During the Learning session, the participants were trained to associate the eight object-object pairs. An identical set of the eight object-object pairs was given to all participants. The Learning session consisted of ten blocks of 16 trials (160 trials in total). During each trial, a cue object was presented for 1-s and a maximum 4-s presentation of two test objects were followed after a 1-s inter-stimulus interval (ISI). For these two objects, the participants were instructed to choose the one associated with the sample object within 4 s. As soon as they responded, feedback ('correct' or 'incorrect') was shown on the screen for 1 s and the next trial then started after a 2-s inter-trial interval (ITI).

The Retrieval session consisted of four runs of 32 trials (128 trials in total). During each trial, a cue object was presented for 500-ms, followed by a 6.5-s ISI and the 300-ms presentation of the test object. The cue object was one of the objects learned during the Learning session. The participants were asked to press the 'yes' button if the cue object and the test object were associated during the Learning session and otherwise to press the 'no' button. They were instructed to respond as quickly as possible within 3 s, and no feedback was given to them.

Throughout the task, the object image size was 384 × 384 pixels, and each object image was presented at a visual angle of 7 ° with a uniform gray background. Between trials, there was a jittered ITI of 0-14 s with an average of 4 s. The order of the objects was randomized and counterbalanced across runs.

To examine whether an active retrieval process is critical for the neural responses, we conducted two additional tasks (passive viewing task and image matching task) in which the participants viewed the learned objects without active retrieval (Figs. S2A and S2C). Due to the limitation of the MRI scan time for ethical reasons, half of the participants (19 participants) performed the passive viewing task and the other 19 participants performed the image matching task after the main task. During each trial of the passive viewing task, a cue object was presented for 500-ms, followed by a 5.5 – 19.5 s ITI. The participants were instructed passively to view each presented object (Fig. S2A). The basic design of the image matching task was identical to the design of the main task except that the cue image could also be presented as a test image and the participants were asked to indicate their judgment if the test object matched the cue object via a button box (Fig. S2C). They were asked to press the 'yes' button if the test object and the cue object were identical and otherwise to press the 'no' button.

A subsequent memory test was conducted at least six months after the main task. The design of the subsequent memory test was identical to the design of the Retrieval session in the main task except that this test was performed in a testing room outside of the scanner. Sixteen of the participants completed this test. Throughout the test, each object was presented eight times as a cue, and there were a total of 128 trials (16 objects × 8 times). The accuracy of the test was calculated as the ratio of correct trials to the total number of trials. An object pair was counted as correct when a participant answered correctly at least seven times for each object in a pair. In the within-subject comparison between the remembered and the forgotten pairs, the data from nine of the participants were used because the other participants remembered (or forgot) just one pair or none.

2.3. Regions-of-Interest

We used both functionally and anatomically defined ROIs. Object-selective and retinotopic early visual regions-of-interest (ROIs) were determined by functional localizer scans. In the object localizer scan, participants viewed alternating 16-second blocks of grayscale object images and retinotopically matched scrambled images (Lee et al., 2012). The resulting object-selective lateral occipital complex (LOC) was used in the analyses. To define the retinotopic regions of the early visual cortex (EVC) corresponding to the central visual field, the participants were asked to view 16-second blocks of a central disk (5°) and an annulus (6-15°) (Lee et al., 2012). Significance maps of the brain were computed by performing a correlation analysis thresholded at a p-value of 0.0001. The mean voxel numbers in LOC and EVC were 2100.47 ± 189.40 and 1334.82 ± 136.59, respectively. The hippocampal regions CA1, CA3, and the dentate gyrus (DG), were automatically defined by means of the tool, which is available in Freesurfer 7 and is based on a statistical atlas built with ultra-high resolution ex vivo MRI data (Iglesias et al., 2015). The mean voxel numbers in CA1, CA3, and DG were 161.89 ± 2.44, 48.76 ± 1.58, and 73.63 ± 1.79, respectively. The regions in the left and right hemispheres were separately analyzed.

2.4. fMRI data acquisition

Participants were scanned on the 3T Siemens MAGNETOM Prisma located at the Center for Neuroscience Imaging Research of the Institute for Basic Science. Images were acquired using a 20-channel head coil with an in-plane resolution of 2 × 2 mm and 35 2 mm slices (0.2 mm inter-slice gap, TR = 2 s, TE = 25ms, field of view = 192 mm, matrix size = 96 × 96). Given the small voxel size, we could only acquire partial volumes of the temporal and occipital cortices. Our slices were oriented approximately parallel to the base of the temporal lobe. All functional localizer and task runs were interleaved. Anatomical image was scanned with MPRAGE sequence.

2.5. fMRI data analyses

The data analyses were conducted using AFNI (http://afni.nimh.nih.gov) (Cox, 1996), SUMA (AFNI surface mapper), FreeSurfer (https://surfer.nmr.mgh.harvard.edu/), and custom MATLAB scripts. Data preprocessing included slice-time correction, motion correction and smoothing. Smoothing was performed only for the localizer data with Gaussian blur of 5 mm full-width half-maximum (FWHM). Then, the percent signal change was calculated by normalization with the average magnitude of the response for each run and each participant on a voxel-by-voxel basis.

To deconvolve the event-related BOLD responses, we utilized a standard general linear model using the AFNI software package (3dDeconvolve using GAM function with motion parameters and up to fourth-order polynomials treated as regressors of no interest). The β-value (in percent signal change) and t-value of each voxel were derived for the peak amplitudes of BOLD responses induced by the events at 0 s (retrieval onset), 2 s, and 4 s after retrieval onset.

In order to examine the representation of associative memory, we determined the degree of pattern similarity for each pair of objects based on a representational similarity analysis (RSA) (Haxby et al., 2001; Kriegeskorte et al., 2008; Lee et al., 2019; Park et al., 2021). Four event-related runs for each participant were divided into two halves in all three possible ways. For each of the splits, cue object-related regressors were modeled using the AFNI 3dDeconvolve with the GAM option as described above. These were modeled separately for each time point (0 s, 2 s, and 4 s after retrieval onset). We estimated the t-value between each event and baseline in each half of the data. The t-values were then extracted from the voxels within each ROI and cross correlated (split-half correlation analysis) (Kravitz et al., 2010; Lee et al., 2012, 2019). The correlation coefficients were Fisher's z transformed for further analyses. The associated pattern index for an ROI was then calculated by subtracting the average of correlation values for the non-learned pairs from the average of correlation values for the learned pairs. During the calculation, the correlations between identical objects were excluded.

In addition to the RSA with split-half correlation analysis described above, we also derived the associated pattern indices based on the trial-by-trial analysis. In this case, the trials of the Retrieval session runs were modeled as separate events, and the estimated t-value for each trial was used to calculate pair-wise Pearson correlations between event-related neural response patterns (trial-by-trial analysis) (Kriegeskorte et al., 2008; Visser et al., 2011, 2013). Then the associated pattern index for an ROI was calculated by subtracting the average of correlation values for the non-learned pairs from the average of correlation values for the learned pairs. During the calculation, the correlations between trials that present the identical object were not included.

To derive discrimination indices for individual objects, we used the split-half correlation analysis (Kravitz et al., 2010; Lee et al., 2012, 2019). The t-values estimated for each half of the data were extracted from the voxels within each ROI and cross correlated. A discrimination index for an object was calculated by subtracting the average of between-condition correlations (Pearson correlation coefficient comparing each object with every other object except for the paired object) from the within-condition correlations (Pearson correlation coefficient comparing each object with other presentations of the same object).

To examine the representational relationship between the visual and hippocampal regions, we used RSA between the ROIs (Haxby et al., 2001; Kriegeskorte et al., 2008; Park et al., 2021). To do this, the trials of the Retrieval session runs were modeled as separate events, and the estimated t-value for each trial was used to calculate pair-wise Pearson correlations between event-related spatial patterns (trial-by-trial analysis) (Visser et al., 2013). The correlation coefficients were Fisher's z transformed for further analyses. Representational dissimilarity matrices (RDMs) containing trial-by-trial dissimilarities between associated objects (LP, learned pairs) or non-associated objects (NP, non-learned pairs) were derived for individual ROIs, after which the correlations between the RDMs of the hippocampal and visual ROIs (LP-LP, NP-NP, LP-NP, NP-LP) were calculated. The correlations between the RDMs of different regions denotes the representational relationships between the regions.

To compare the object correlation values during retrieval for the remembered pairs with those for the forgotten pairs at the time of the subsequent memory test performed at least six months after retrieval, we used the t-values derived from the aforementioned trial-by-trial analysis. We calculated the average of the correlation values for the remembered pairs and the average of the correlation values for the forgotten pairs for each participant.

2.6. Statistical analyses

The associated pattern indices or representational relationships were compared with basal level (zero) by two-tailed one-sample t-tests. To compare object correlation values, or representational relationships, we used two-tailed paired-samples t-tests. To determine statistical significance in any effect of a time point, ROI, or the interaction between them, repeated-measures ANOVAs (tests of within-subjects effects) were conducted. For all of the ANOVAs with factors more than two levels, Greenhouse-Geisser corrections were used if sphericity assumptions were not met. To examine the contribution of different learned and non-learned pair numbers to the pattern similarity, bootstrap analyses were conducted. To do this, we randomly selected eight pairs from the non-learned pairs and derived object correlation values for the selected non-learned pairs. We repeated these steps 10,000 times and tested whether the average of the object correlation values from the learned pairs fell within the top 5% of the simulated null distribution of the object correlation value for the non-learned pairs. We compared different pair types (LP-LP, LP-NP, NP-LP, NP-NP) in representational relationships between different ROIs after a false discovery rate (FDR) multiple comparisons correction. We used SPSS to conduct the statistical analyses.

3. Results

3.1. Learning of object associations

Prior to the Learning session, the participants rated the relatedness of the objects by means of a Likert scale (1; not related -5; highly related). Results revealed a low score (Fig. S1) indicating that our stimuli for the main task was reliable. During the Learning session, participants were trained to associate eight fixed object pairs (learned pairs) (Figs. 1A, 1B, and S1A). On the last block of the Learning session, the participants showed 98.85 ± 0.46% accuracy. All participants also showed high accuracy in the Retrieval session for the object pairs that they learned in the Learning session (95.37 ± 0.86%) (Fig. 1C), indicating successful learning and retrieval of associative memory.

3.2. Reinstatement of associative memory traces in the hippocampal and cortical regions

To investigate the reinstatement of associative memory traces in the hippocampus during retrieval, we initially focused on the neural response patterns at the time of retrieval onset in the hippocampal regions CA1, CA3, and DG. The representations of associative object memory were examined by comparing the degree of pattern similarity for object pairs. Specifically, we extracted the responses in each ROI in two independent halves of the data, and calculated the correlations across the two halves for the object pairs that were associated (learned pairs) during learning and for the pairs that were not associated (non-learned pairs) (split-half correlation analysis) (see Methods) (Table S1). Then we compared the degree of pattern similarity between the learned and non-learned pairs (Fig. 2A). If associative memory traces that contain shared features acquired during the Learning session are represented in the hippocampal region, the pattern similarity of the learned pairs may be greater than that of the non-learned pairs in the region. We found that in the left CA1, learned pair correlations were significantly greater than non-learned pair correlations at the time of retrieval onset ($t = 2.167$, $p = 0.037$) (Fig. 2B). However, in CA3 or DG of the left hemisphere or hippocampal regions of the right hemisphere, significantly greater pattern similarity for the learned pairs than the non-learned pairs were not observed (Figs. 2B and S3A). Thus, these results show the representation of associative memory traces in CA1, suggesting memory reinstatement in CA1 at the time of retrieval onset.

Consistent with the result of CA1, in the visual cortical regions, lateral occipital complex (LOC) and early visual cortex (EVC) in the left hemisphere, we also found significantly greater pattern similarity for the learned pairs compared to the non-learned pairs at retrieval onset (lLOC, $t = 2.383$, $p = 0.022$; lEVC, $t = 2.269$, $p = 0.029$) (Fig. 2C), indicating that associative memory traces are reinstated in the sensory cortex. For the visual cortex, the same tendency was also observed in the right hemisphere (rLOC, $t = 2.050$, $p = 0.048$; rEVC, $t = 3.263$, $p = 0.002$) (Fig. S3C). Taken together, these results suggest that the hippocampal region CA1 with the visual cortex is involved in memory reinstatement, which is immediately induced as soon as a retrieval cue is given.

3.3. Changes in the representations of associative memory in the hippocampal regions

We subsequently examined whether the neural representations of associative memory traces change after retrieval onset in the hippocampal regions. Based on the response patterns evoked at retrieval onset and two- and four-seconds following retrieval onset, we derived associated pattern indices as the difference between the learned pair correlations and the non-learned pair correlations. Because CA3 and DG showed the same tendency of pattern change across the time points, we combined the subregions into CA3/DG to increase the power.

Notably, we found the opposite tendency in the change of the associated pattern indices across time points from retrieval onset between CA1 and CA3/DG in the left hippocampus (Fig. 2D). A two-way ANOVA with time points (TP: 0, 4 s) and ROI (CA1, CA3/DG) as factors revealed significant interaction between TP and ROI ($F(1,37) = 8.728$, $p = 0.005$) (Fig. 2D). In the left CA3/DG, significant associated patterns were observed at 2 s and 4 s but not at 0 s (retrieval onset) ($t = 0.706$, $p = 0.485$ at 0 s, $t = 2.119$, $p = 0.041$ at 2 s, and $t = 4.542$, $p = 5.744 \times 10^{-5}$ at 4 s) and there was a significant increase in associated pattern indices from retrieval onset to 4 s ($t = 2.123$, $p = 0.040$) (Fig. 2D). In contrast, there was a significant associated pattern at retrieval onset but not at later time points in the left CA1 ($t = 2.167$, $p = 0.037$ at 0 s, $t = 1.447$, $p = 0.156$ at 2 s, and $t = 0.295$, $p = 0.769$ at 4 s) (Fig. 2D). We additionally derived the associated pattern indices based on the trial-by-trial analysis (see Methods) (Table S1) and found the same tendency observed in the associated pattern indices derived from the split-half correlation analysis (Fig. S4).

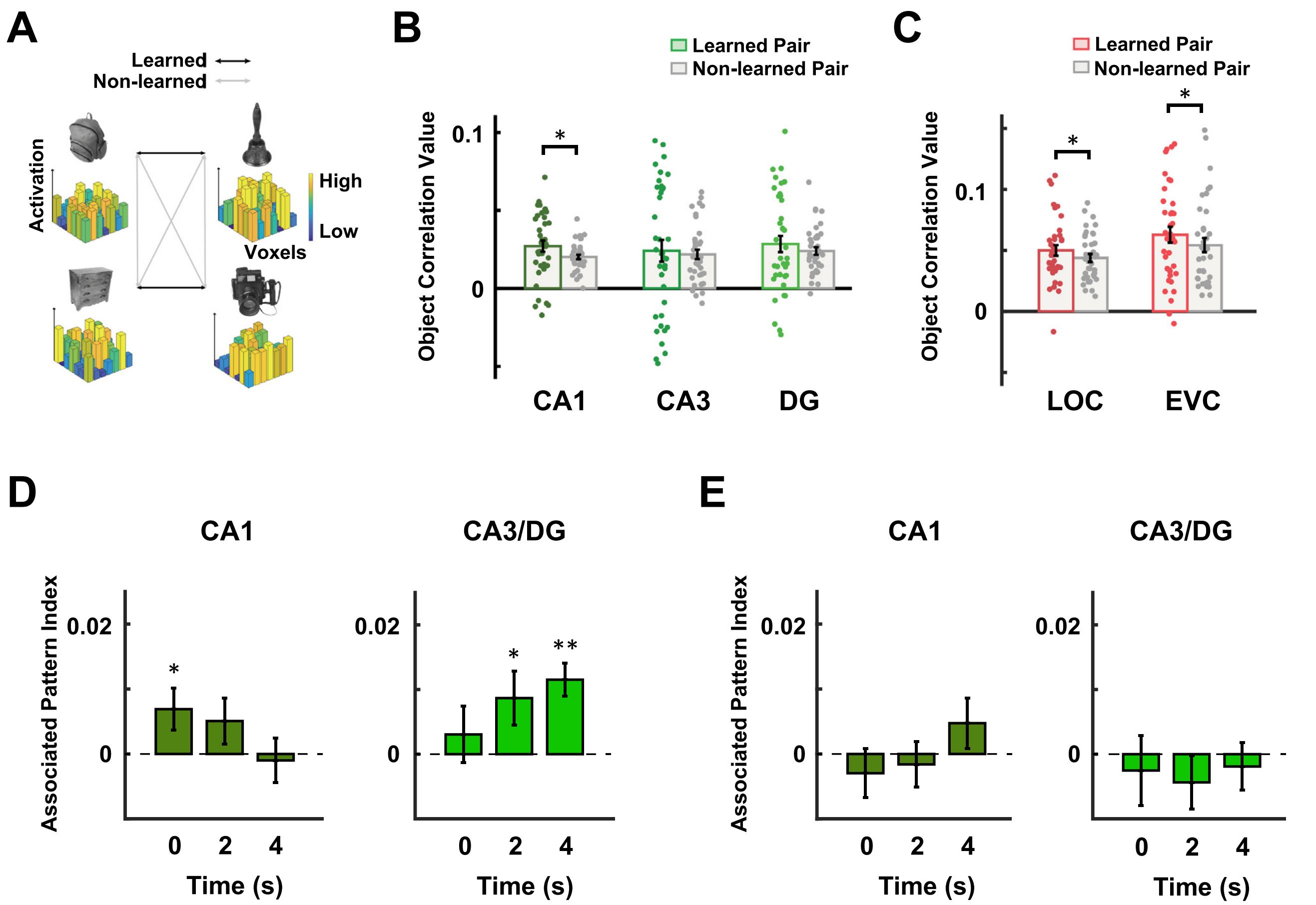

Fig. 2. Representations of associative memory during retrieval. (A) Schematic description of the calculation of the pattern similarity for object pairs. The bar graphs depict the neural response pattern evoked for a specific object, with each bar in the graph denoting the magnitude of the BOLD response in each voxel. Correlations between neural patterns of the learned pairs and those of the non-learned pairs were calculated and compared. **(B)** The degree of pattern similarity for the learned pairs and the non-learned pairs in the hippocampal subfields CA1, CA3, and DG at retrieval onset. CA1 showed significantly higher correlations between the learned pairs than the non-learned pairs. $^*p < 0.05$. Error bars indicate between-subjects s.e.m and each dot indicates mean correlation value for each participant. **(C)** The degree of pattern similarity for the learned pairs and the non-learned pairs in the visual areas LOC and EVC at retrieval onset. The LOC and the EVC showed significantly higher correlations between the learned pairs than the non-learned pairs. $^*p < 0.05$. Error bars indicate between-subjects s.e.m and each dot indicates mean correlation value for each participant. **(D)** Changes of the associated pattern indices over time during retrieval in CA1 and CA3/DG. While CA1 showed significant representation of associative memory at retrieval onset and weaker representations at subsequent time points, CA3/DG showed gradually stronger representations of associative memory over time during retrieval. $^*p < 0.05$, $^{**}p < 0.01$, compared to the basal level (the degree of pattern similarity for the non-learned pairs) (dotted line). The data at 0s are the difference scores of the data in (B). Significant interaction effect between time points (TP: 0, 4 s) and ROI (CA1, CA3/DG) was revealed by two-way repeated measures ANOVA ($F(1,37) = 8.728$, $p = 0.005$). Error bars indicate between-subjects s.e.m. **(E)** Changes of the associated pattern indices in CA1 and the CA3/DG over time during the follow-up experiment in which the same participants viewed the learned objects without active retrieval. No significant changes in the associated pattern indices were found. All areas here are areas of the left hemisphere. Error bars indicate between-subjects s.e.m. The dotted lines indicate the basal levels of associated patten indices.

Given the absence of significant representation in the right hippocampal regions (Fig. S3B), CA1, CA3/DG, and the visual regions henceforth indicate the regions in the left hemisphere unless otherwise specified. These results show different temporal dynamics in associated patterns between CA1 and CA3/DG, indicating different information processing between the hippocampal regions from retrieval onset.

Because there are more non-learned pairs than learned pairs, we examined in more depth whether the difference between the numbers of pairs contributes to the difference in the associated patterns between CA1 and CA3/DG. To do this, we randomly selected non-learned pairs equal in number to the learned pairs, and calculated the pattern similarities and associated pattern indices. By repeating this procedure 10,000 times, we found that the pattern similarity between the learned pairs is still significantly greater than that of the non-learned pairs in CA1 at 0 s ($p = 0.040$), and in CA3/DG at 2 s ($p = 0.019$) and 4 s ($p = 0.002$) (Fig. S5), indicating little contribution of the different numbers of pairs.

We also examined the magnitude of responses during retrieval in CA1 and CA3/DG. We found decreasing tendency over time in both CA1 and CA3/DG, and no relationship between the associated pattern indices and the BOLD magnitudes (Fig. S6).

To confirm whether the dynamic change of the associated patterns in CA1 and CA3/DG depends on the retrieval signals, we conducted a follow-up fMRI experiment in which the same participants viewed the learned objects without active retrieval (see Methods) (Figs. 2E and S2). We performed the same analysis on the associated pattern index and found no significant changes in the associated pattern indices in both CA1 and CA3/DG. A two-way ANOVA with time points (TP: 0, 4 s) and ROI (CA1, CA3/DG) as factors revealed no main effects or interactions involving TP (all $F(1,37) < 1.298$, $p > 0.262$) (Fig. 2E). During these follow-up tasks, as well as the Retrieval session of the main task, normal visual responses were preserved, allowing us to decode individual objects from the response patterns of the visual cortex (Fig. S7). These

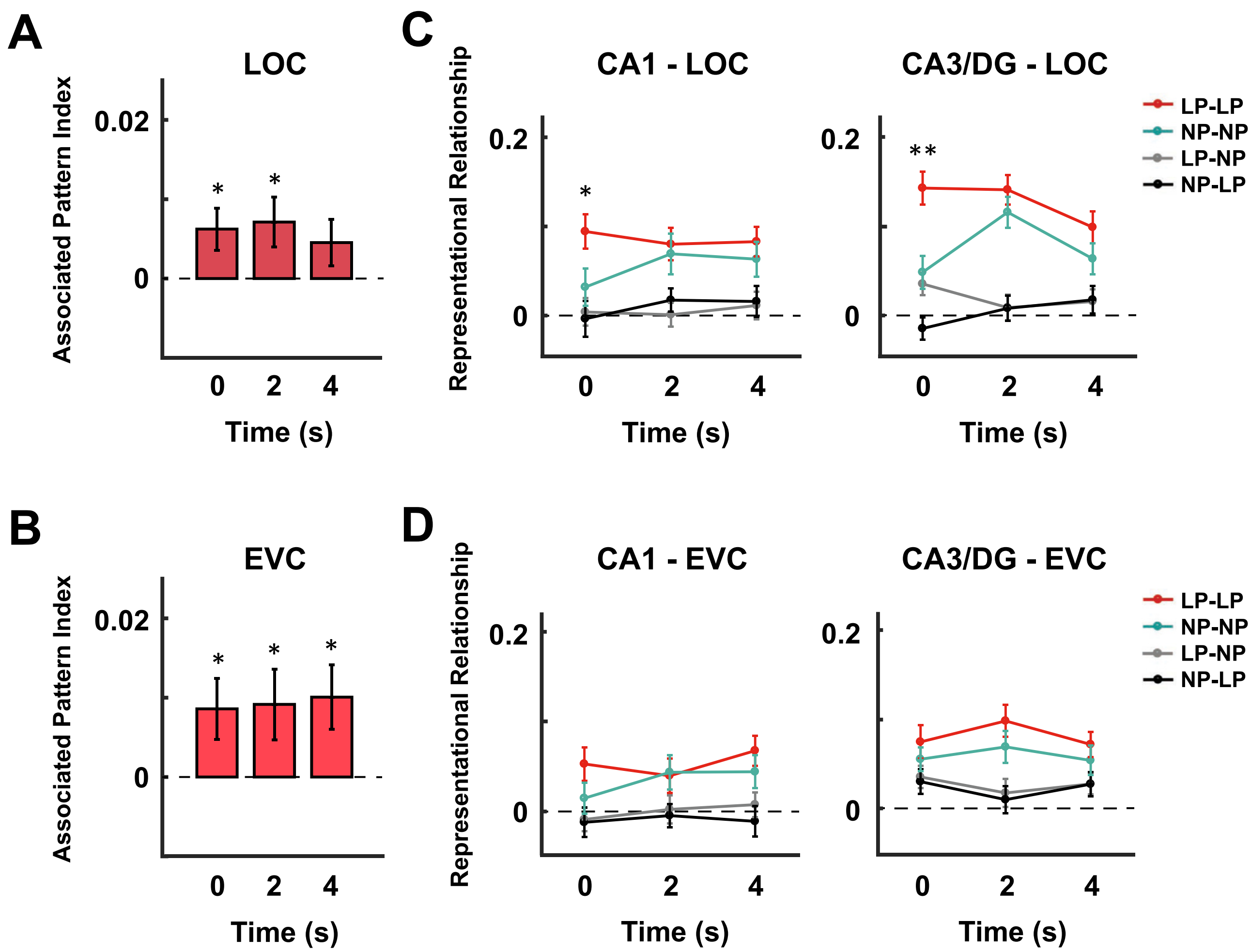

Fig. 3. Representational relationship between the visual and hippocampal regions during retrieval. **(A)** Changes of the associated pattern indices over time during retrieval in the LOC. $^{*}p < 0.05$ compared to the basal level (dotted line). Error bars indicate between-subjects s.e.m. **(B)** Changes of the associated pattern indices over time during retrieval in the EVC. $^{*}p < 0.05$ compared to the basal level (dotted line). Error bars indicate between-subjects s.e.m. **(C)** Representational relationship between CA1 and the LOC (left panel) and relationship between CA3/DG and the LOC (right panel) over time during retrieval. The degrees of representational similarity for the learned pairs (LP-LP), the non-learned pairs (NP-NP), and cross pairs (LP-NP or NP-LP) were derived and compared. A one-way ANOVA revealed significant effects of pair types for the comparison between CA1 and the LOC ($F(3,111) = 6.791$, $p = 3.020 \times 10^{-4}$) and for the comparison between CA3/DG and the LOC ($F(2.187,80.922) = 23.721$, $p = 2.692 \times 10^{-9}$) at retrieval onset. Significant representational relationships between CA1 and LOC, and between CA3/DG and LOC were found for the learned pairs (LP-LP) compared to other pair types ($^{*}p < 0.05$, $^{**}p < 0.01$, FDR corrected). Error bars indicate between-subjects s.e.m. **(D)** Representational relationship between CA1 and the EVC (left panel) and relationship between CA3/DG and the EVC (right panel) over time during retrieval. Error bars indicate between-subjects s.e.m.

results suggest that the associated pattern change observed in CA1 and CA3/DG during retrieval is triggered by retrieval signals.

3.4. Hippocampal processing beyond the reinstatement

In our results, the representation of associative memory was observed in CA1 and CA3 sequentially for four seconds after retrieval onset (Fig. 2D). However, these directions of information and duration were quite different from those previously reported in other studies of reinstatement (Granger et al., 1996; Rolls, 1996). Moreover, the increase in the associated pattern in CA3/DG was observed after cortical reinstatement (Fig. 2D). Therefore, the gradual change of the CA3/DG representations in our results suggests the possibility that there is a different type of information processing, such as the strengthening of memory traces triggered by retrieval signals beyond reinstatement.

To confirm the existence of information processing beyond reinstatement, we examined the changes in pattern responses in the visual cortex across time points and the relationship between the visual and hippocampal representations (Fig. 3). The significant associated patterns observed at retrieval onset in the LOC and EVC were maintained throughout the time points, except at 4 s for the LOC (LOC: $t = 2.383$, $p = 0.022$ at 0 s; $t = 2.289$, $p = 0.028$ at 2 s; $t = 1.555$, $p = 0.128$ at 4 s; EVC: $t = 2.269$, $p = 0.029$ at 0 s; $t = 2.079$, $p = 0.045$ at 2 s; $t = 2.518$, $p = 0.016$ at 4 s) (Figs. 3A and 3B). These results suggest that the associative memory traces in the visual cortex are also represented immediately after retrieval onset, consistent with the result observed in CA1, and that the reinstated memory traces in the visual cortex are maintained throughout the retrieval.

To reveal whether the visual representations of memory traces correspond to the hippocampal representations, we subsequently examined the representational relationship between the visual and hippocampal regions based on a representational similarity analysis (RSA) (Haxby et al., 2001; Kriegeskorte et al., 2008; Park et al., 2021). To do this, representational dissimilarity matrices (RDMs) containing trial-by-trial dissimilarities between associated objects (LP, learned pairs) or non-associated objects (NP, non-learned pairs) were derived for individual ROIs. The correlations between the RDMs of the hippocampal and visual ROIs were then calculated for the learned pairs (LP-

LP), the non-learned pairs (NP-NP), and the cross pairs (LP-NP, NP-LP) (see Methods). Remarkably, one-way ANOVA revealed significant effect of pair types (LP-LP, NP-NP, LP-NP, NP-LP) ($F(3,111) = 6.791$, $p = 3.020 \times 10^{-4}$) for the comparison between CA1 and LOC at retrieval onset (0 s) (Fig. 3C). We found that the degrees of representational similarity for the learned pairs (LP-LP) between CA1 and LOC were significantly greater than those for the non-learned pairs (NP-NP) ($t = 2.171$, $p = 0.036$, FDR corrected) or those for the cross pairs ($t = 3.293$, $p = 0.002$ compared to LP-NP; $t = 3.293$, $p = 0.002$ compared to NP-LP, FDR corrected) at retrieval onset. In the comparison between CA3/DG and LOC at retrieval onset, significant effect of pair types was also found ($F(2.187,80.922) = 23.721$, $p = 2.692 \times 10^{-9}$, one-way ANOVA) (Fig. 3C). Additionally, the degrees of representational similarity for the learned pairs (LP-LP) between CA3/DG and LOC were also significantly greater than those for the non-learned pairs (NP-NP) ($t = 3.622$, $p = 0.001$, FDR corrected) or those for the cross pairs ($t = 6.518$, $p = 1.255 \times 10^{-7}$ compared to LP-NP; $t = 7.063$, $p = 2.340 \times 10^{-8}$ compared to NP-LP, FDR corrected) at retrieval onset. This tendency in the representational relationship was observed only at retrieval onset but not at subsequent time points (Fig. 3C). Thus, the increase of the associated pattern indices observed in CA3/DG at later time points during retrieval may reflect information processing beyond reinstatement, such as memory strengthening, that is induced by retrieval signals.

Additionally, no significant representational relationship between the EVC and hippocampal regions was found despite the significant representation of the associative memory throughout the measured time points (Figs. 3B and 3D). These results suggest that visual reinstatement occurs immediately after retrieval onset through the interaction between the hippocampus and the high-level visual cortex.

If the increase in the associated pattern indices in CA3/DG after memory reinstatement (Fig. 2D) reflects the memory strengthening process triggered by retrieval signals, subsequent memory performance may depend on the change of the associated pattern in CA3/DG during retrieval. To test this possibility, we asked the participants to perform a subsequent memory test at least six months after retrieval. The design of the subsequent memory test was identical to the design of the Retrieval session in the main task. Sixteen of the participants completed this test and the accuracy level was $78.467 \pm 4.735\%$ (Fig. S8). We counted an object pair is correct when a participant answered correctly at least seven times of total eight repetitions for each object in a pair. Based on this criterion, the mean number of correct pairs was 4.125 ± 0.679 out of a total of eight pairs (Fig. S8).

We next directly compared the object correlation values during retrieval between the remembered pairs and the forgotten pairs at the time of the subsequent memory test. In this analysis, we excluded the data from the participants who remembered (or forgot) just one pair or none. The nine participants included showed the accuracy level of $77.257 \pm 2.472\%$ and the mean number of correct pairs was 3.222 ± 0.344 (Fig. S8). Notably, we found significantly different tendency in the change of object correlation values across time points from retrieval onset between the remembered pairs and the forgotten pairs in CA3/DG (Fig. 4A). A two-way ANOVA with time points (TP: 0, 4 s) and the pairs (remembered, forgotten) as factors revealed significant interaction between TP and the pair ($F(1,8) = 7.283$, $p = 0.027$) (Fig. 4A). Moreover, the object correlation values for the remembered pairs at 4 s after retrieval onset were significantly greater than those at retrieval onset in CA3/DG ($t = 3.186$, $p = 0.013$) while this tendency was not observed for the forgotten pairs ($t = 0.291$, $p = 0.779$) (Fig. 4A). Furthermore, the object correlation values for the remembered pairs were significantly greater than those for the forgotten pairs in CA3/DG at 4 s after retrieval onset ($t = 3.534$, $p = 0.008$) (Fig. 4A). In CA1, no significant relationship between the object correlation values and the subsequent memory performance was observed. For remembered pairs, a two-way ANOVA with time points (TP: 0, 4 s) and the ROI (CA1, CA3/DG) as factors revealed significant interaction between the TP and the ROI ($F(1,8) = 15.117$, $p = 0.005$) (Fig. 4A). This tendency was not observed for the forgotten pairs ($F(1,8) = 0.082$, $p = 0.781$). In the object correlation values for the non-learned pairs, no significant effect was found throughout retrieval (Fig. 4B). These results provide evidence supporting the existence of the retrieval-induced strengthening of memory traces in CA3/DG.

4. Discussion

Our findings suggest the existence of hippocampal processing beyond the reinstatement of memory traces during retrieval. While CA1 showed significant representation of associative memory at retrieval onset and weaker representations at subsequent time points, CA3/DG showed gradually stronger representations of associative memory over time during retrieval. Moreover, the correspondence between the hippocampal and visual representations was observed at retrieval onset but not at the following time points for both CA1 and CA3/DG. These results indicate that the gradual increase in the degree of associative memory over time in CA3/DG may reflect a strengthening process during retrieval rather than the reinstatement process. Furthermore, this processing observed in CA3/DG was related to the subsequent long-term memory performance measured at six months or later. Collectively, our findings suggest the existence of a retrieval-induced strengthening process of associative memory trace in CA3/DG, underlying the long-lasting maintenance of memory.

4.1. Information processing during retrieval beyond reinstatement in CA3/DG

Our results show gradually stronger representations of associative memory over time during retrieval in CA3/DG (Fig. 2D). Given the short time range of memory reinstatement processing in our results (Fig. 3A), as well as in prior research (Jang et al., 2017; Staresina et al., 2012b, 2016; Yaffe et al., 2014), we examined the possibility that this gradual increase in the strength of associative memory in CA3/DG may reflect retrieval-induced neural processing beyond memory reinstatement. While the reinstatement of stored memory traces is critical for successful memory retrieval (deBettencourt et al., 2019; St-Laurent et al., 2015; Tanaka et al., 2014), prior research suggests that memory retrieval itself can act as a signal to trigger strengthening or modifications of the retrieved memory traces (Rossato et al., 2006, 2010). If such strengthening processing is triggered from the retrieval cue onset, the processing may be observed in parallel with the reinstatement processing even during retrieval. Our result showing correspondence between the hippocampal regions and the LOC at retrieval onset, but not at the following time points (Fig. 3C), support the possibility that the observed stronger representations of associative memory in CA3/DG after retrieval onset, reflect beyond-reinstatement processing such as strengthening. Moreover, the subsequent memory test performed at least six months after retrieval support that the stronger representations of associative memory in CA3/DG after reinstatement are associated with the strength of subsequent memory (Fig. 4). The proposed retrieval-induced strengthening of associative memory in CA3/DG was not observed when a cue object was just viewed without active retrieval (Fig. 2E), indicating that active retrieval is required to trigger the strengthening process in CA3/DG. These results collectively suggest that there exists a form of retrieval-triggered neural processing beyond reinstatement, which can contribute to strengthening memory traces in CA3/DG.

Prior studies suggest that activation of the hippocampal traces is critical for cortical reinstatement of memory (Teyler and DiScenna, 1986; Teyler and Rudy, 2007). The activity of the hippocampus was correlated with stimulus-specific reinstatement in the visual cortex during cued recall (Bosch et al., 2014; Tanaka et al., 2014) and the representation of reinstated memory was also found in the hippocampus during retrieval (Backus et al., 2016; Chadwick et al., 2010; Dimsdale-Zucker et al., 2018; Hsieh et al., 2014). Our results extend these findings by showing significant representations of associative memory in CA1 as well as the

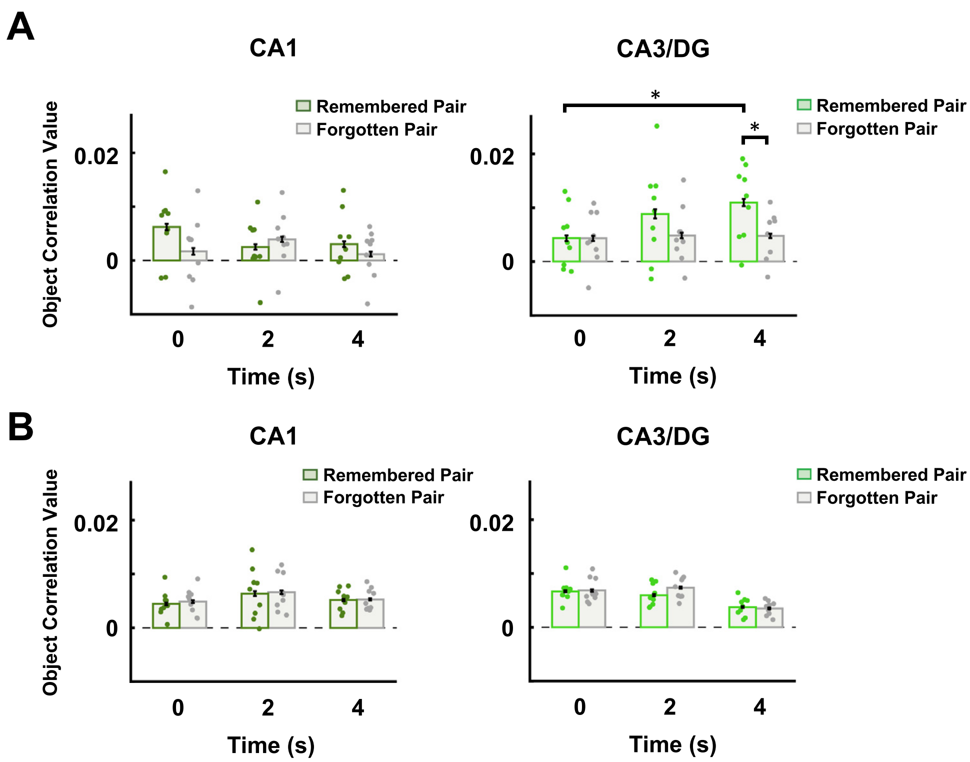

Fig. 4. Associative memory representations in the hippocampus during retrieval for remembered and forgotten pairs at the subsequent memory test. (A) The degree of pattern similarity in CA1 (left panel) and CA3/DG (right panel) during retrieval for remembered pairs and forgotten pairs based on the subsequent memory performance. While CA1 did not show significant difference between remembered and forgotten pairs, CA3/DG showed significantly higher pattern similarity for remembered pairs than forgotten pairs 4 s after retrieval onset (*$p < 0.05$). Moreover, the object correlation values for the remembered pairs at 4 s were significantly greater than those at retrieval onset in CA3/DG (*$p < 0.05$). Error bars indicate between-subjects s.e.m and each dot indicates mean correlation value for each participant. **(B)** The degree of pattern similarity in CA1 (left panel) and CA3/DG (right panel) during retrieval for the non-learned pairs. Error bars indicate between-subjects s.e.m and each dot indicates mean correlation value for each participant.

LOC at the time of retrieval onset (Figs. 2B and 2C). Moreover, significant correspondence between the representations of CA1 and the LOC was found at retrieval onset (Fig. 3C). This representational correspondence was also found between CA3/DG and the LOC despite the absence of statistically significant representation of associative memory at retrieval onset in CA3/DG (Figs. 2D and 3C). The unavailable detection of associative memory at retrieval onset in CA3/DG may reflect the sparse activity induced by the retrieval cue in CA3/DG (Marr, 1971; Poli et al., 2017; Viskontas et al., 2006; Wixted et al., 2014). Thus, these results, combined with the finding of the CA3/DG strengthening process at later time points, suggest that CA3/DG plays a dual role during retrieval for both memory reinstatement and strengthening.

The recurrent excitatory connections of CA3 have been considered to play a critical role in mnemonic processes such as association, cued recall, and pattern completion (Bennett et al., 1994; Gilbert and Brushfield, 2009; Gluck and Myers, 1997; Granger et al., 1996; Kesner and Rolls, 2015; Rolls, 1996; Treves and Rolls, 1992, 1994). The recurrent architecture of CA3 should be advantageous for rapid encoding and retrieval of associative memories (Haberman et al., 2008; Kesner, 2007; Kesner and Rolls, 2015; Rebola et al., 2017). Thus, based on these circuits, CA3 could be a fast player leading the strengthening triggered by retrieval. Our data do not mean that CA1 is not involved in the retrieval-induced memory strengthening process. The fast strengthening-process in CA3 may affect CA1 processing and the interaction between CA1 and cortical regions for systems consolidation, which involves reorganization of the hippocampal and cortical circuits that encode the memory over longer time durations, such as days, weeks, months, or even years (Barry and Maguire, 2019; Buzsáki, 1996; Dudai, 2004; Kitamura and Inokuchi, 2014; Remodes and Schuman, 2004). Additionally, if the retrieval signal is sufficient to trigger the strengthening process in CA3, the same effect may be observed for the retrieval of remote memories that formed months or years before retrieval. These possibilities require additional investigations in future work.

While both the left and right hemispheres of the visual cortex were used to represent associative memory, significant representation of associative memory was found only in the left hippocampal regions (Figs. 2D, 3A, 3B, S3B and S3D). This lateralization of hippocampal processing in our results could reflect the specific property of the stimuli we used in the tasks. In earlier work, the effect of left hippocampal sclerosis on an associative verbal memory task compared to that of right hippocampal sclerosis was prominent (Saling et al., 1993). Complementary representation in the left and right hippocampus was also reported during navigation during which place information was represented in the right hippocampus, while temporal sequences were observed in the left hippocampus (Iglói et al., 2010). Moreover, Motley and Brock Kirwan, showed that pattern separation could be observed in the left hippocampus during semantic information processing, while pattern separation of spatial information critically involved the right hippocampus (Motley and Brock Kirwan, 2012). Although our stimuli were object images, it is still possible that the participants in our experiment used a language-based strategy to associate objects, or that associative memory was semantically processed. Additionally, the studies focusing on the similarity between the representations during encoding and retrieval show predominant involvement of the right hippocampus (Mack and Preston, 2016; Tompary et al., 2016). The representations during retrieval in the left hippocampus in our results could emphasize more sematic properties than the representations during encoding. Furthermore, while many prior studies used short-term memory paradigms, in which a retrieval test was performed just after learning, we used a long-term memory paradigm, in which the retrieval session was given one day after the learning session. Over the first day following learning, hippocampal representations could undergo a significant reorganization (consolidation)(Lee et al., 2019). Thus, the difference in paradigm can also induce the discrepancy in the lateralization of hippocampal processing between our results and those in other studies. Further research will be needed to investigate the nature of representations in the left or right hippocampus during retrieval.

4.2. Reinstatement during retrieval in the cortical areas

We found the representation of associative memory in the visual cortex during retrieval. We suggest that this visual representation reflects the reinstatement of visual information evoked by activated hippocampal traces, given the following evidences. First, we found significant representation of associative memory at retrieval onset in the visual cortical regions as well as in the hippocampal region of CA1 (Figs. 2B and 2C). Second, there was significant correspondence between the representations of the LOC and hippocampal regions at retrieval onset (Fig. 3C). Finally, prior research and memory models suggest that the activation of a hippocampal memory trace can induce cortical reinstatement of the encoded memory (Rugg and Vilberg, 2013; Woodruff et al., 2005). Furthermore, our results of visual representation during retrieval also suggest that the following two points are worthy of consideration. First, while both the LOC and EVC represent associative memory during retrieval, representational correspondence with the hippocampal regions was only observed in the LOC and not in the EVC (Fig. 3). This result implies that the LOC more closely interacts with the hippocampus for the reinstatement of memory and contains a structure of information more similar to that of the hippocampus than to that of the EVC. The reinstatement in the EVC may be induced by signals conveyed from the LOC, but the representation of the EVC may reflect different aspects of visual information (such as low-level aspects) from those processed in the LOC. Second, correspondence between the representations of the LOC and the hippocampal regions was observed only at retrieval onset but not during the following times, while significant associative memory was represented in the visual cortex across the entire duration of retrieval (Figs. 3A and 3C). This result may indicate that once the visual reinstatement is activated through interaction with the hippocampus, subsequent processes occur independently of the hippocampus within the visual regions.

It may also be possible that not only maintenance of the reinstated information but also additional information processing, such as generalization or categorization, proceeds at later time points following retrieval onset. The relatively high level of representational relationship for non-learned pairs (NP-NP) as well as learned pairs (LP-LP) at later time points (Fig. 3C) might reflect such additional processing. This possibility needs to be explored in future works. Additionally, retrieval-induced strengthening process of memory might proceed even during and after the presentation of the test object. In this study, it was difficult to confirm this suggestive finding because, given the temporal resolution of fMRI data and the present task design, the strengthening process could not be distinguished from other cognitive processes including test object perception and motor action planning. Further studies based on the new design considering the limitation in temporal resolution may be needed to address this.

5. Conclusion

In conclusion, we show differential information processing in CA1 and CA3/DG during memory retrieval. Our results suggest the existence of neural processing beyond the reinstatement in CA3/DG during retrieval. This neural processing may underlie the memory strengthening triggered by retrieval signals, providing an explanation of how retrieval itself can improve human memory performance.

Data and code availability statement

The behavioral and neuroimaging data and codes are available from the corresponding author on reasonable request. Sharing and reuse of data require the expressed written permission of the authors, as well as clearance from the Institutional Review Boards.

Credit author statement

Joonyoung Kang: Methodology, Software, Validation, Formal analysis, Investigation, Writing - Original Draft, Writing - Review & Editing, Visualization. **Wonjun Kang:** Writing - Review & Editing, Visualization. **Sue-Hyun Lee:** Conceptualization, Methodology, Resources, Writing - Original Draft, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare no competing interests.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2022.119493.

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